



Slides 5.16: Plan and Prototype

Lesson 5.16, one day (day 137). Eleven slides. Pins and scissors are out today, so the count-out and count-in run.

Slide 1: Do now, and count out

- Look at your material and your brief.
- Write the two measurements you have to get right.
- Then write what happens if you get one of them wrong by an inch.
- **Station Managers: count the pins out loud and write the number in today's row on the station count sheet.**

Notes: hold the one-minute conferences with the “needs a conference” briefs while the rest write. Hook: hold up a piece cut two inches too short next to a finished pencil case. This one was going to be that. Nobody cuts fabric today until I initial a plan.

Slide 2: Measure twice, cut once

- Say it back to me.
- You cannot uncut fabric.
- Ten minutes of planning and one paper mock-up saves the whole project.

Slide 3: The three marks on a plan

- **Cut line:** a solid line
- **Fold line:** a dashed line
- **Stitch line:** a dotted line
- Use all three on your sketch.

Image: one flat pattern piece drawn with all three line types labeled.

Slide 4: Finished size and cut size

- **Finished size** is how big the thing is when it is done.
- **Cut size** is bigger. The difference is the **seam allowance**.
- Seam allowance for this project: **half an inch on every edge you sew.**
- Example: a pouch finished at 8 by 4 inches, sewn on three edges, is cut at about 9 by 4.5 inches.
- **Check: is your cut size bigger than your finished size? If not, fix it now.**

Notes: this is the number one reason an item comes out too small. Ask every student for both numbers at the approval table.

Slide 5: The order of work

- mark → cut → fold → pin or clip → stitch the structural seams → attach the closure → finish the raw edges → knot and trim → test with the real contents
- **Order matters.** You cannot pin after you stitch. You cannot cut after you sew.
- Which step can you not undo?

Slide 6: Build the mock-up of the tricky part

- Not the whole item. **One part.** At full size. Out of paper.
- The part you wrote on your brief as most likely to go wrong.

- Then hold it against the real material and check that it fits.

Notes: circulate with one question per student: show me the part you were worried about.

Slide 7: The tricky part, by path

Path	Mock this up
T-shirt tote	Where the handle meets the body
Headband	The length around a real head, plus the overlap
Phone sleeve	The closed end, and whether the phone still fits
Drawstring bag	The channel: how wide for the cord to pull
Patchwork square	The corner where three pieces meet
Pencil case	The flap length and where the button lands
Pillow cover	The corner where two seams meet
Produce bag	The top channel, and the towel's thickness
Hand warmer	The opening: how much you can fill and still close it
Wristlet	The inside pocket, and whether your hand fits through

Slide 8: The approval table. Five questions.

1. Is the material a used textile?
2. Are your two stitches two of the four taught?
3. Does your cut size include the seam allowance?
4. Is your order of work actually possible?
5. Is your reduced-scope version written?
 - Bring the plan sheet, the mock-up, and the material.
 - **Fabric shears are at my table. Nothing is cut before my initials are on your plan.**

Slide 9: While you wait for approval

- Write your order of work in final form.
- Mark your fabric with chalk. Do not cut.
- Read the peer review checklist so you know what your partner will ask.
- Check Rubric 05 criterion 1 against your own plan sheet.

Slide 10: Peer review. Two stars and a step.

- Trade sheets with your station partner.
- **Two things that will work.** Name the part, not “it’s good.”
- **One change to try**, with a reason.

- Then on your own sheet: the change I am taking, and the change I am not taking and why.
- Banned: “it’s good.” “I like it.” “Nothing to change.”

Slide 11: Count in, then closure

- **Every pin back in the cushion. Station Managers count out loud and write the second number. All three counts must match before anybody is dismissed.**
- Scissors back in the kit, blades closed.
- Closure line: tomorrow I start by _____. The part I am worried about is _____.
- Everything in the labeled bag: plan sheet, mock-up, material, brief.

Notes: the count comes first. Nobody writes until all three match. If one is short, everybody stops and looks.